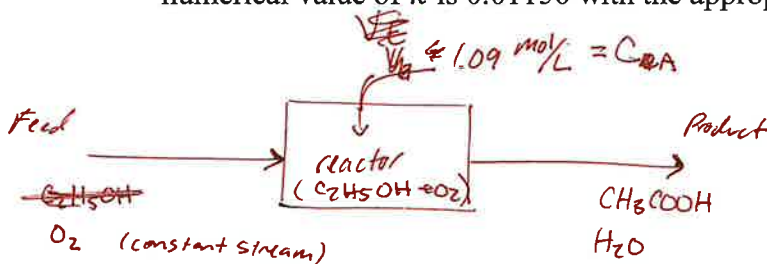


To Receive Full Credit, You Must Show All Your Work

Ethanol is converted to acetic acid by the reaction $C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$. The process takes place in a batch reactor initially filled with an aqueous solution of ethanol with concentration 1.09 mol/L. A stream of O_2 is passed through the reactor to maintain a constant concentration of dissolved O_2 in the reactor throughout the process. The reaction rate is second order in the ethanol concentration: $r_A = kC_A^2V$ where $C_A = n_A/V$ and V is assumed to be constant at 2500 L. The subscript A indicates the ethyl alcohol. Ignore any volume change caused by the production of water.

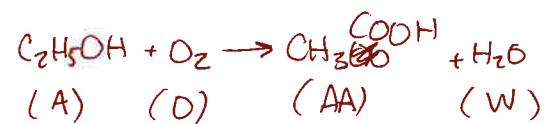
- (a) If the reaction rate r_A has units of mol/day, and C_A has units of mol/L, what are units for the constant k ?
 (b) Write a mole balance for ethanol. Combine this with the reaction rate.
 (c) Use the result of part (b) to calculate the time to convert 85% of the ethanol to acetic acid. The numerical value of k is 0.01156 with the appropriate units from part (a).



$$V = 2500 \text{ L}$$

$$r_A = k C_A^2 V$$

$$C_A = \frac{n_A}{V}$$



(a)

$$r_A [=] \frac{\text{mol}}{\text{day}}$$

$$C_A [=] \frac{\text{mol}}{\text{L}}$$

$$V [=] \text{L}$$

$$k [=] ?$$

$$r_A = k C_A^2 V$$

$$\frac{\text{mol}}{\text{day}} = k \left(\frac{\text{mol}}{\text{L}} \right)^2 (\text{L})$$

$$\frac{\text{mol}}{\text{day}} = k \frac{\text{mol}^2}{\text{L}}$$

$$\frac{\text{mol}}{\text{day}} = k \frac{\text{mol}^2}{\text{L}} \rightarrow \frac{\text{mol}}{\text{day}} \cdot \frac{\text{L}}{\text{mol}^2} = k$$

$$k [=] \text{L/mol-day}$$

(b) mole balance Ethanol

Accumulation = In - out + generated - consumed

$$\text{Acc.} = I_n - O_n = 0 \quad (\text{assume steady state})$$

$$0 = n_{A, \text{in}} - n_{A, \text{out}} \quad \text{if } n_{A, \text{in}} = n_{A, \text{out}} \rightarrow n_{A, \text{out}} = n_{A, \text{in}}$$

$$C_A = \frac{n_A}{V} \rightarrow C_A V = n_A \rightarrow \left(\frac{1.09 \text{ mol A}}{\text{L}} \right) (2500 \text{ L}) = n_{A,0} = 2725 \text{ mol A} \quad \checkmark$$

$$n_{A, \text{out}} = n_{A,0}$$

$$\text{So, } C_{A, \text{out}} = C_{A,0}$$

$$r_A = k C_A^2 V$$

$$r_A' = 2k V C_A = \frac{dC_A}{dt}$$

$$\frac{dC_A}{dC_A} = 2k V / C_A$$

$$\frac{dC_A}{C_A} = 2k V dt$$

$$\int \frac{dC_A}{C_A} = \int 2k V dt \rightarrow \ln(C_A) = (2k V)t + C$$

$$C_A = C_0 e^{-(2k V)t}$$

@ $t=0$, $C_A(0) = 1.09 \text{ mol/L}$

$$\rightarrow C_A(0) = C_0 e^{-2k V(0)}$$

$$1.09 \text{ mol/L} = C_0$$

$$C_A(t) = 1.09 e^{-2k(2500)t}$$

NAME _____

Some useful integrals:

$$\int (a + bx) dx = \frac{1}{2b} (a + bx)^2$$

$$\int \frac{dx}{(a + bx)} = \frac{1}{b} \ln|a + bx|$$

$$\int \frac{dx}{x(a + bx)} = -\frac{1}{a} \ln \left| \frac{a + bx}{x} \right|$$

$$\int \frac{x dx}{(a + bx)} = \frac{1}{b^2} (a + bx - a \ln|a + bx|)$$

$$\int \frac{dx}{(a + bx)^2} = \frac{-1}{b(a + bx)}$$

$$\int \frac{(A + Bx)}{(a + bx)} dx = \frac{(bA - aB) \ln|a + bx| + bBx}{b^2}$$

$$\int \frac{dx}{(a + bx)(A + Bx)} = \frac{1}{(bA - aB)} \ln \left| \frac{a + bx}{A + Bx} \right|$$

① $k = 0.01156$ 85% consumed NAME ADISON KENNEDY

85% ethanol to acetic acid (mol ratio is 1:1) → how long to reach 85%, so we want to equate $C_A(t)$ to $(1-0.85)(1.09 \text{ mol/L})$
 $C_A(t) = 1.09 e^{-2(2500)kt}$ $t = ? \text{ days}$
 ~~$= 0.1690 = 0.1643 \text{ mol/L}$~~

$$\rightarrow \frac{0.1643}{1.09} = \frac{1.09 e^{-2(2500)(0.01156)t}}{1.09}$$

$$\rightarrow 0.15 = e^{-57.8t}$$

$$\ln(0.15) = \ln(e^{-57.8t})$$

$$\frac{-1.897119}{-57.8} = \frac{-57.8t}{-57.8}$$

$$t = 0.03282 \text{ days or } 47.26 \text{ minutes}$$

② $r_A = k C_A^2 V$ $C_A = \frac{n_A}{V} \rightarrow C_A V = n_A = 2725 \text{ mol A}$

$$\rightarrow r_A' = 2k V C_A^2$$

$$\rightarrow r_A' = 2k n_A = \frac{dn_A}{dt} = -r_A$$

$$\frac{dn_A}{dt} = 2k n_A$$

$$\frac{dn_A}{n_A} = 2k dt$$

$$\int \frac{dn_A}{n_A} = \int 2k dt$$

$$\ln(n_A) = 2kt + C$$

$$\ln(n_A) = e^{2kt} + e^C$$

$$n_A = C e^{2kt}$$

$$n_A @ t=0 \Rightarrow 2725 \text{ mol A}$$

$$\rightarrow 2725 \text{ mol A} = C e^{-2k(0)}$$

$$2725 = C$$

$$\rightarrow n_A(t) = 2725 e^{-2kt}$$

ok
 This is correct for 1st-order Rx, I think these are correct but Rx is 2nd order

③

$$(0.85)(2725) = 2316.25 \text{ mol A consumed}$$

$$408.75 = 2725 e^{-2kt}$$

$$0.15 = e^{-2kt}$$

$$\ln(0.15) = \ln(e^{-2(0.01156)t})$$

$$-1.8971 = -0.02312 t$$

$$t = 82.055$$

$$2316.25 = 2725 e^{-2kt}$$

$$0.85 = e^{-2(0.01156)t}$$

$$-0.1625 = -0.02312 t$$

$$t = 7.029 \text{ days}$$

units do not work

